

Title of Topic: FUNCTIONS IN SQL

Course Code: 23IT201

Course Title: DataBase Management Systems

MODULE –II

Academic Year: 2024 - 2025 (Even Sem)

MAX

```
SELECT MAX(age)  
FROM Customers;
```

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

SELECT MAX(age)
FROM Customers;

MAX(age)
31

MIN

```
SELECT MIN(age)  
FROM Customers;
```

ALIAS NAME

```
SELECT MAX(age) AS max_age  
FROM Customers;
```

MIN IN STRINGS

--select the minimum value of first_name from Customers

```
SELECT MIN(first_name) AS min_first_name  
FROM Customers;
```

MIN() function in a nested SELECT statement

```
SELECT *  
FROM Customers  
WHERE age = (  
    SELECT MIN(age)  
    FROM Customers  
);
```

HAVING CLAUSE

```
SELECT *
```

```
FROM Customers
```

```
GROUP BY country
```

```
HAVING MAX(age);
```


LIMIT

```
SELECT first_name, age  
FROM Customers  
LIMIT 2;
```

LIMIT,OFFSET

- **LIMIT 2** selects two results
- **OFFSET 3** excludes the first three results

```
SELECT first_name, last_name  
FROM Customers  
LIMIT 2 OFFSET 3;
```

AVG()

--selects the average amount spent by each customer from the Orders table

```
SELECT customer_id, AVG(amount) AS average_spends  
FROM Orders  
GROUP BY customer_id;
```

SUM()

--select the sum of the amount of id 4 from orders

```
SELECT SUM(amount) AS total_of_cus4  
FROM Orders  
WHERE customer_id = 4;
```

GROUPBY

-- select the item column and the count of order ids from the Orders table

-- group them by the item column

```
SELECT COUNT(order_id), item
```

```
FROM Orders
```

```
GROUP BY item;
```

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

SELECT country, COUNT(*) AS number
FROM Customers
GROUP BY country;

country	number
UAE	1
UK	2
USA	2

HAVING

- The SQL HAVING clause is used if we need to filter the result set based on aggregate functions such as MIN() and MAX(), SUM() and AVG(), and COUNT().

```
SELECT AggFunc(column), extra_columns  
FROM table  
GROUP BY target_column  
HAVING condition
```

HAVING

-- select the count of customer ids greater than one and their corresponding country

```
SELECT COUNT(customer_id), country  
FROM Customers  
GROUP BY country  
HAVING COUNT(customer_id) > 1;
```


Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

`SELECT COUNT(customer_id), country
FROM Customers
GROUP BY country
HAVING COUNT(customer_id) > 1;`

COUNT(customer_id)	country
2	UK
2	USA

HAVING Vs WHERE CLAUSE

HAVING Clause

The HAVING clause checks the condition on **a group of rows**.

HAVING is used with aggregate functions.

The HAVING clause is executed after the GROUP BY clause.

WHERE Clause

The WHERE clause checks the condition **on each individual row**.

The WHERE clause cannot be used with aggregate functions.

The WHERE clause is executed before the GROUP BY clause.

ORDERBY

-- orders all rows from Customers in ascending order by country

SELECT *

FROM Customers

ORDER BY country;

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

SELECT *
FROM Customers
ORDER BY first_name;

customer_id	first_name	last_name	age	country
5	Betty	Doe	28	UAE
3	David	Robinson	22	UK
1	John	Doe	31	USA
4	John	Reinhardt	25	UK
2	Robert	Luna	22	USA

ORDERBY ASC

-- orders all rows from Customers in ascending order by age

SELECT *

FROM Customers

ORDER BY age ASC;

ORDERBY DESC

-- order all rows from Customers in descending order by age

SELECT *

FROM Customers

ORDER BY age DESC;

DISTINCT

-- select the unique countries from the customers table

```
SELECT DISTINCT country  
FROM Customers;
```

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE



```
SELECT DISTINCT country  
FROM Customers;
```



country
USA
UK
UAE

COUNT

```
SELECT COUNT(DISTINCT country)  
FROM Customers;
```

COUNT

--returns the number of rows in the Customers table

```
SELECT COUNT(*)
```

```
FROM Customers;
```